

LANDON BAKKEN

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Education

University of Wisconsin Madison

Bachelor in Computer Engineering, expected July 2028

Sept 2024 – Present

Madison, Wisconsin

Applicable Classes: ECE 252, 352, 210, 230 354 — CS 300, 400

Experience

Wisconsin Racing Formula SAE

Sept 2024 – Present

Controls Lead

Madison, Wisconsin

2026 Electric Car

- Software, Driver, and Physics in loop simulation that contributed to **1st place Design**, reducing on-track tuning/development time by 64.4%, and allowed for 64% more control systems developed
- Reduction Traction Control **3rd place Autocross** and **5nd place Efficiency**
- Speed Estimation **3nd place Autocross**
- Traction Control **1nd place Acceleration**
- Created a wireless telemetry system

2025 Electric Car

- Developed and tuned traction control with load transfer feedforward and slip error feedback controller - reduced laptimes resulted in **2nd place Autocross**
- Developed and tuned torque vectoring based on speed and steering angle, and was well integrated with traction control. Recognized by the **Multimatic Vehicle Dynamics Award**
- Developed driver interface relating charge and distance left in endurance that was specifically recognized for **1st place in Overall Design**
- Managed a multi-bus CAN communication system

Combustion Car

- Developed systems to tune engine cylinder air to fuel ratio, resulting in a 13.2% power increase, and **3rd place in Efficiency**
- Made torque model that allows for precise and immediate torque cuts for traction control, faster pneumatic shifting, and faster throttle response, contributing to **3rd place in Overall Design**

Mercury Marine

May 2026 – Present

Embedded Controls Engineer

Oshkosh, WI

- Developed a GNS program with demo features to reduce design cycle time
- Hull simulator visualizer to accelerate development of new virtual anti-collide system
- Removed sensor noise from booster pump overshoot, reduced peak-to-peak noise by 45%

UW Makerspace

Sept 2024 – Present

Technical Staff

Madison, WI

- Assist students with operating tools and equipment at the UW Madison Makerspace
- Trained students on how to safely use composites room for carbon fiber work

Personal Projects

Machine Learning | *Neural Networks, Gradient Descent, Python*

Dec 2024 – Jan 2025

- Built a machine learning model and training system from scratch using gradient descent in Python

Multiplayer Networking | *Networking, C#, Unity, Documentation*

Nov 2023 – Dec 2024

- Developed a low-latency, peer-to-peer multiplayer system for Unity using UDP, TCP, and HTTP protocols, with comprehensive documentation for users without networking experience

3D Engine on a Calculator | *Low-level programming, Extreme Optimization*

Oct 2021 – May 2023

- Created a 3D game engine on a TI-84+ CE using raycasting, a technique used in early DOOM games

Technical Skills

Computer Languages: Simulink, Python, C#, Java, C++, Javascript, Verilog, CSS/HTML

Manufacturing: SLA and FDM 3D printing, Laser Cutting, Soldering, CNC

Development Tools: Git, VS Code, Kvaser, Unity

3D Design: Blender, Fusion, Solidworks

Concepts: PID Controllers, CAN Protocol, Traction Systems, Signal Filtering, Networking